8th

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A priori error estimation.

$$\begin{cases} -(au')' = f \text{ on } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

with P_1 FEM

(1) energy norm:

$$\|u - U\|_E^2 = \int_0^1 a[(u-U)']^2 dx$$

Here, $\|u - U\|_E \le \|u - v\|_E$ for $\forall\ v \in V_h$

take $v = \pi_h u$

$$\|e\|_E \le C \|h u''\|_a$$

(2) L^2 norm:

$$\|u - U\|^2 = \int_0^1 (u-U)^2 dx$$

Summary

$$\|e\| \le C_I^2 h_{max}^2 \|u''\|_a \frac{\|\phi''\|_a}{\|e\|}$$

Duel problem:

$$-\phi'' = e \qquad \phi(0) = \phi(1) = 0$$

User problem

$$\left(-a\phi'\right)' = e \quad , \quad \phi(0) = \phi(1) = 0$$

~~claim~~ claim

$$\begin{cases} -(a\psi')' = \xi \\ \psi(0) = \psi(1) = 0 \end{cases}$$

Then $$\max_{\xi \neq 0} \frac{\|\psi''\|_a}{\|\xi\|} \leq c_s$$

where c_s only depends on

proof

$$-a\psi'' = a'\psi' + \xi$$

$$\|a\psi''\| \leq \|\xi\| + \|a'\psi'\|$$

$$\leq \|\xi\| + \|a'\|_\infty \|\psi'\|$$

$$\hookrightarrow \|a''\|_\infty = \max_{x \in [0,1]} |a'(x)|$$

Take BVP & multiply by ψ and int-by-parts

$$-\int_0^1 \psi\,(a\psi')'\,dx = \int_0^1 \xi\,\psi\,dx$$

$$= \int_0^1 a(\psi')^2 dx = \int_0^1 \zeta \psi \, dx$$

$$a_{\min} \int_0^1 (\psi')^2 dx \le \int_0^1 \zeta \psi \, dx$$

$$\therefore a_{\min} \|\psi'\|^2 \le \|\zeta\| \, \|\psi\|$$

Aside: Poincaré inequality, let $v \in V$, then

$$v(x) = \int_0^x v'(y) \, dy$$

$$|v| \le \int_0^1 |v'| \, dy$$

$$\le \sqrt{\int_0^1 (v')^2 \, dy}$$

$$\therefore \int_0^1 v^2 dx \le \int_0^1 (v')^2 dy$$

$$\|v\| \le \|v'\|$$

$$\|v\| \leq \|v'\|$$

Back to proof:

$$a_{min} \|\psi'\|^2 \leq \|\xi\| \|\psi\|$$

$$\leq \|\xi\| \|\psi'\| \quad \text{— by Poincaré}$$

$$\therefore \|\psi'\| \leq \frac{\|\xi\|}{a_{min}}$$

Now, $\|a\psi''\| \leq \|\xi\| + \|a'\|_\alpha \|\psi''\|$

~~Now~~, Then

$$\|a\psi''\| \leq \|\xi\| + \|a'\|_\alpha \frac{\|\xi\|}{a_{min}}$$

$$\|a\psi''\| \leq \left(1 + \frac{\|a'\|_\alpha}{a_{min}}\right) \|\xi\|$$

but $\sqrt{a_{min}} \|\psi''\|_a \leq \|a\psi''\|$

$$\left(\|\psi''\|_a = \sqrt{\int_0^1 a(\psi'')^2 dx}\right.$$

$$\text{but } \|a\psi''\| = \sqrt{\int_0^1 a^2 (\psi'')^2 dx}$$

$$\therefore \|\psi''\|_a \leq \underbrace{\left(\frac{1 + \frac{\|a''\|_\infty}{a_{min}}}{\sqrt{a_{min}}} \right)}_{c_s} \|\xi\|$$

$$\therefore \frac{\|\psi''\|_a}{\|\xi\|} \leq c_s$$

$$\therefore \|e\| \leq \underbrace{c_I^{\nu} c_s}_{\text{constant}} \|h^{\nu} u''\|_a$$

— or —

$$\|e\| \leq \underbrace{c_I^{\nu} c_s h_{max}^{\nu}}_{\text{constant}} \|u''\|_a$$

∴ assuming $\|u''\|_a$ is bounded,

then $\|e\| = O(h^{\nu})$

A posteriori error

(1) energy norm:

$$\|e\|_E = \|e'\|_a$$

$$\|e\|_E^2 = \int_0^1 a(e')^2 dx$$

$$= \int_0^1 a u' e' dx - \int_0^1 a v' e' dx$$

$$= \int_0^1 f e dx - \int_0^1 a v' e' dx$$

subtract from this:

$$\int_0^1 f v dx - \int_0^1 a v' v' dx = 0$$

$$u/v = \pi_h e \in V_h$$

$$\|e\|_E^2 = \int_0^1 f(e - \pi_h e) dx - \int_0^1 a v' (e - \pi_h e)' dx$$